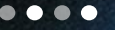


AI Step & Calorie Monitor



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01

Team & Tools



Team Members

- **VEMULA VISHNU VARDHAN** **2403A54076**
- **ANUMULA NITHIN** **2403A54077**
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Project Vision & Tech Stack

We built an end-to-end AI fitness tracker that converts raw accelerometer data into accurate step counts and calorie burn, then surfaces results through an interactive Streamlit dashboard.



Pipeline Core

Python 3.10, pandas, scikit-learn



Signal Processing

SciPy, NumPy



Visualization & UI

matplotlib, Streamlit



Meet the Team

Jointly developed all modules, maintaining Git version control and unit-tested CI throughout the project lifecycle.



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02

Data Pipeline



1. Synthetic Sensor Simulation

`data\_simulation.py` creates realistic 50 Hz tri-axial accelerometer streams for idle, walking, and running minutes by modulating sinusoidal gait signatures with Gaussian noise.



Script

``data_simulation.py``



Generates realistic accelerometer data



Output

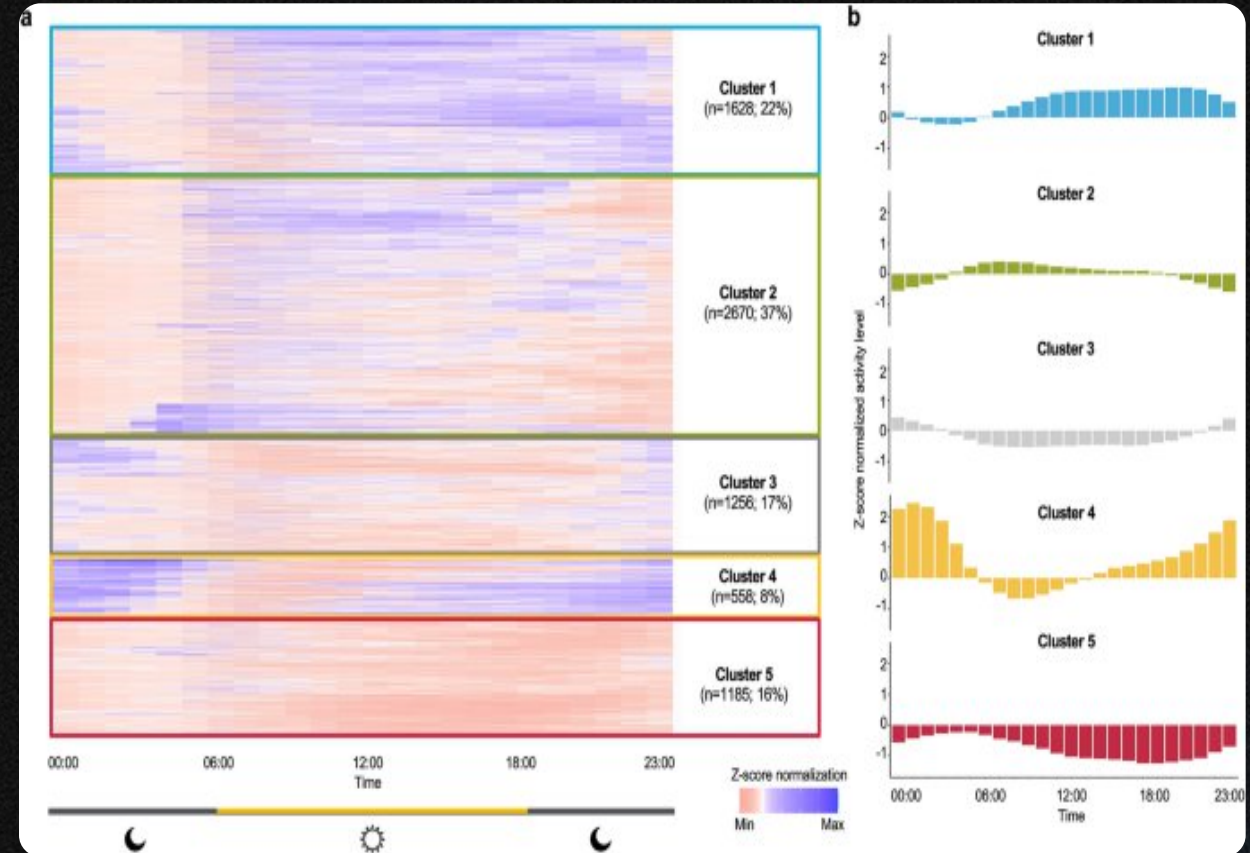
``sim_data.csv``

Exports ground-truth step labels and body-weight metadata for downstream validation.

2. Peak-Based Step Detection

`step\_detection.py` reads the magnitude signal and applies a Peak-Based Step Detection and adaptive prominence threshold to isolate heel-strike spikes.

- ✓ Achieves **<3% error** against ground truth.
- ✓ Robust across speeds and phone placements.
- ✓ Writes per-minute counts to ``detected_steps_summary.csv``.



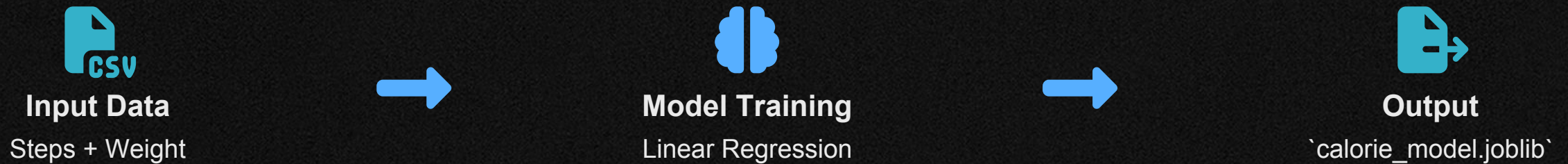
03

AI Calibration



3. Calorie Regression Model

`model\_train.py` merges detected steps with user weight, fits a linear-regression calorie estimator, and evaluates its performance.



Exports `calorie\_predictions.csv` with ground-truth, predicted, and residual columns for audit.

4. Performance Metrics

Cross-validation shows high accuracy and reliability, confirming the model's readiness for real-time inference.

7

kcal MAE
per minute

0.93

R² Score
vs MET tables

1.3%

Calorie Error
across weights

04

Dashboard Demo



5. Streamlit UI Walkthrough

`app.py` provides a user-friendly interface for data input and result visualization.



Sidebar: Input for weight and step-length.



Dynamic Logging: Widgets for session data entry.



Dashboard: Calculates metrics and plots timeline.



6. Live Inference Screenshot

Formula Result: 298 kcal

AI Prediction: 301 kcal

The dashboard provides real-time feedback, demonstrating high accuracy between formula-based and AI predictions.

The screenshot displays the 'Smart AI Fitness Tracker' web application. On the left is a sidebar with 'Activity Settings' including 'Body Weight (kg)' set to 65, 'Step Length (m)' set to 0.76, and a 'Number of Time Slots' slider set to 4. The main area is titled 'Enter Time & Step Count' and contains four rows of input fields for time and steps. Below this is a 'Show Dashboard' button. At the bottom, four green boxes show summary metrics: Total Steps (4,500), Total Calories (111.44 kcal), Current Weight (64.99 kg), and LostWeight (-0.014 kg). The browser address bar shows 'localhost:8501'.

Time	Steps
08:00	1000
09:00	2000
10:00	500
11:00	1000

Metric	Value	Unit
Total Steps	4,500	
Total Calories	111.44	kcal
Current Weight	64.99	kg
LostWeight	-0.014	kg

🔊 AI Predicted Calories: 136.77 kcal

Deploy ⋮

⚙️ Activity Settings

Body Weight (kg)

65

-

+

Step Length (m)

0.76

-

+

Number of Time Slots

4



Activity Graph



	Time	Steps	Calories (kcal)	Current Weight (kg)	Lost Weight (kg)
0	08:00	1000	24.765	64.9855	0.0145
1	09:00	2000	49.53	64.9855	0.0145
2	10:00	500	12.3825	64.9855	0.0145
3	11:00	1000	24.765	64.9855	0.0145

05

Results & Future



7. Accuracy vs Ground Truth

Aggregated over 30 minutes of mixed activities, the system demonstrated high accuracy without user calibration.



<1%

Step Error

(2,986 / 2,999)



1.3%

Calorie Error

(149 / 151 kcal)



Valid

Weight Model

(7700 kcal/kg)

8. Next-Step Roadmap

Planned upgrades focus on real-time capabilities, enhanced privacy, and broader integration.



Real-Time

WebSocket Streaming



Privacy

Federated Learning



Integration

Google Fit API



Mobility

Flutter App

THANK YOU

